
Compositional Concept Synthesis via CLIP-Guided Primitive Stitching

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Abstract

In this paper we explore the task of creating a composite image given a natural language prompt from a set of given primitives, such as combining “person” and “umbrella” to form “a person under an umbrella.” Each primitive is represented as a visual element, and a composition function arranges them spatially to form a complete scene. To guide this composition, we use CLIP to evaluate how well the rasterized image of the composed scene aligns with a natural language description of the target concept. The optimization objective is to maximize the cosine similarity between the image embedding and the text embedding produced by CLIP.

1. Introduction

Our work focuses on the synthesis of complex composite image by stitching together a set of predefined primitives. Each target concept is associated with a collection of k primitives, which are arranged to maximize the cosine similarity between the generated image embedding and the corresponding text embedding produced by CLIP. Initially, we implemented this task using the `pgpelib` library, where each concept was directly mapped to its corresponding primitives.

Early experiments revealed that our initial optimization methods were insufficient, leading us to explore alternative strategies. We discovered that the Covariance Matrix Adaptation (CMA) algorithm performed slightly better.

Despite the improvements provided by CMA, the performance of our method was hindered by a challenging loss function landscape. The loss, defined as the negative cosine similarity, exhibited many local maxima, which impeded the optimization process from converging to the global optimum. To address this, we modified the loss function by

incorporating an additional component designed to smooth the landscape. This adjustment aimed to mitigate the effect of local maxima, thereby facilitating more robust convergence.

While this enhanced approach yielded to better results, it also introduced certain limitations that will be discussed in detail in the following sections.

2. Related Work

We based our work on the paper *Modern Evolution Strategies for Creativity: Fitting Concrete Images and Abstract Concepts* (Tian & Ha, 2022) (ES-CLIP).

ES-CLIP presents a method that leverages evolutionary strategies to optimize image generation based on CLIP’s joint embedding space. An element in the *population* is composed of a set of overlapping triangles eachone with its own vertex position and color. These features are optimized to maximize the cosine similarity between the CLIP embedding of the triangles rendered on the canvas and the CLIP embedding of a target prompt.

Inspired by this work, we want to do something similar by composing visual scenes from a set of semantic primitives, instead of triangles.

3. Method

In order to stitch together k primitives to form a scene, we need to define a few components: the parameter to be optimized and the optimization strategy.

3.1. Parameters & Pipeline

We represent the parameters as a matrix of shape $(k, 5)$ and each entry is bounded to be in the $[0, 1]$ range. Each row corresponds to one primitive and the five parameters describe its spatial attributes.

- **x, y**: the position of the top-right corner of the primitive with respect to the canvas size;
- **scale**: the relative size of the primitive (i.e. a percentage of the full primitive size);

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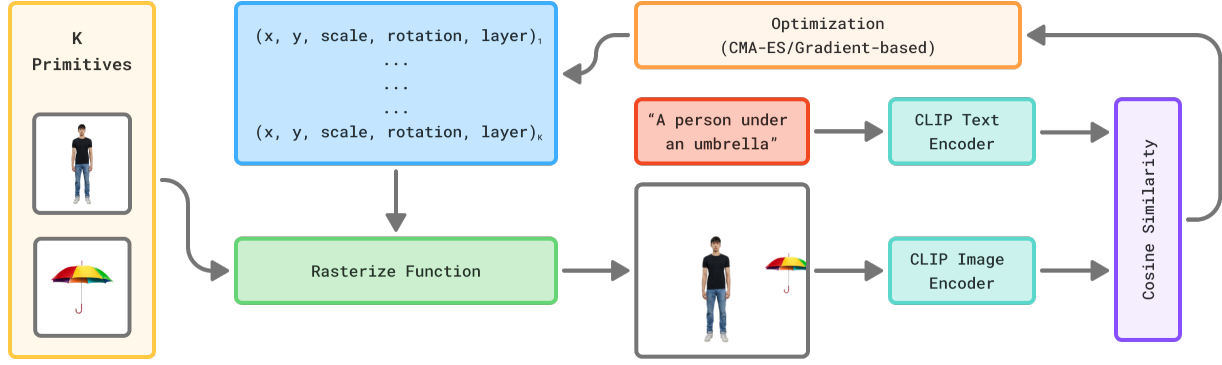


Figure 1. The whole pipeline of our method. We want the k (in this case 2) primitives shown in the yellow box, to be rasterized to have a similar CLIP embedding as the prompt, shown in the red box.

- **rotation**: the angle of rotation $r \in [0, 365]$ degrees normalized;
- **layer**: the rendering order (i.e., which primitives appear in front). We order the rasterization process, such that primitives with smaller layer parameter will be rasterized first, hence be in the back.

We created a `rasterize` function, which uses these parameters and the primitives to compose the scene and rasterizes it into an image. This image is then fed into CLIP’s image encoder to compute its embedding. The cosine similarity between this image embedding and the text embedding of the target prompt serves as the objective function. An overview of this pipeline is shown in Figure 1.

3.2. PGPE Optimization

The first algorithm we tried is PGPE (Probabilistic Gradient Population Evolution) (Toklu et al., 2020), the one used in the ES-CLIP paper. This algorithm is a gradient-free optimization method that uses a population of candidate solutions to explore the search space. This method did not yield good results as in the implementation of the ES-CLIP paper.

3.2.1. PGPE PROBLEM

We hypothesized that one of the reason can be found in the fact that in our problem the solutions have very **sparse rewards**. It means that sampled points in the surroundings of a certain point in the solution space yields inconsistent rewards values. If the best solution is far from the sampled points, the algorithm will not be able to reach it, since intermediate solutions do not provide an incremental improvement in the reward. It means it is difficult for the algorithm to find a direction to move in the search space.

3.3. CMA-ES Optimization

To overcome the limitations of PGPE, we decided to use a more advanced Evolution Strategy implementation, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Hansen & Ostermeier, 1996). Compared to PGPE, CMA-ES uses a more sophisticated sampling algorithm, which prioritize directions where improvement has been observed updating a multivariate Gaussian distribution. Unfortunately even if it produces slightly better results compared to PGPE, this method do not brings a significant improvement.

3.4. Gradient-Based Approach

In parallel, we also experimented with a gradient-based optimization method. We created a new rasterization function which is differentiable and has some limitations: rotating the primitives was pretty hard, so we decided to implment it only if a preliminary test would have been promising. This function allowed us to compute gradients of the cosine similarity loss with respect to the primitive parameters and apply gradient descent to improve the layout. Unfortunately it do not yielded good results as well.

3.4.1. THE CLIP PROBLEM

Given the fact that both the gradient-based and the CMA-ES methods did not yield good results, we decided to investigate the problem further.

3.5. Loss Function Exploration

After many experiments we found out that both methods often became stuck in poor local optima, so we decided to explore the loss function’s landscape. This eventually led us to explore modifications to the loss in an effort to change the optimization surface, as detailed in the next section.

4. Results

In this section we will report the results of each method individually, then some common improvements we applied to both of them and finally we will show the loss exploration we talked about in the previous section.

4.1. Default approach

This approach is the one we implemented first, using the CMA-ES python library. For the gradient-based approach we just implemented the `differentiable_rasterize` function (which operates on PyTorch tensors) and used the Adam optimizer to update the parameters of the primitives. This yielded poor results as shown in the figure.

4.2. Loss Exploration

After these poor results with both methods, we decided to explore the loss function landscape. We found that the loss function was very noisy and had many local optima, which made it difficult for the optimization algorithms to converge to a good solution. An example is shown in the figure. In order to plot the contour plot of the loss function, we choose a simple example "A person under an umbrella", that has two primitives: *person* and *umbrella*. We embedded every combination of $(x, y, scale)$ parameters with values 0.3, 0.38, 0.46, 0.54, 0.62, 0.7 for both primitives, while keeping the rotation and layer fixed. The loss function was computed for each combination and plotted in a contour plot. Of course, close points in this space are not guaranteed to be close in the original space, but still this is useful because it shows us how the majority of parameters combination yield the same cosine similarity.

4.3. Loss Improvements

5. Discussion and Conclusions

TODO

References

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